

¹
² Plasticity, nutrition, and breeding
³

⁴ **Fecundity as a linear function**

First, we model external conditions as a normal distribution

$$C(y) = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right)$$

And fecundity as a linear function:

$$F(n, y) = a_f n + cy.$$

Then, the fitness is given by:

$$W(n, y) = C(y)F(n, y) = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) (a_f n + cy)$$

To find the optimal breeding date, we take the partial derivative against y :

$$\frac{\partial W(n, y)}{\partial y} = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) \times (c - y(a_f n + cy)/w^2)$$

Setting this equal to zero, we have:

$$y = \frac{-a_f n + \sqrt{(a_f n)^2 + 4c^2 w^2}}{2c}$$

Now, if we average the fitness $W(n, y)$ over the breeding date distribution y , we

get the mean fitness:

$$\begin{aligned}
\bar{W}(n, \bar{y}) &= \int p(y)W(n, y)dy \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma_y^2}\right) \times \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{(y-\mu)^2}{2\sigma_y^2}\right) \times \exp\left(-\frac{y^2}{2w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{w^2(y-\mu)^2 + \sigma_y^2 y^2}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{w^2(y^2 - 2\mu y + \mu^2) + \sigma_y^2 y^2}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{(w^2 + \sigma_y^2)y^2 - 2w^2\mu y + w^2\mu^2}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{(w^2 + \sigma_y^2)y^2 - 2w^2\mu y}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{y^2 - 2w^2\mu y/(w^2 + \sigma_y^2) + (w^2\mu/(w^2 + \sigma_y^2))^2 - (w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2 - (w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \exp\left(\frac{w^2\mu^2}{2\sigma_y^2(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2\sigma_y^2}\right) \exp\left(\frac{w^2\mu^2}{2\sigma_y^2(w^2 + \sigma_y^2)}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy
\end{aligned}$$

We still need to spend more time doing the integrals:

$$\begin{aligned}
\bar{W}(n, \bar{y}) &= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \\
&\times \frac{\sqrt{2\pi\sigma_y^2 w^2/(w^2 + \sigma_y^2)}}{\sqrt{2\pi\sigma_y^2 w^2/(w^2 + \sigma_y^2)}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\times (a_f n + cy) dy \\
&= \frac{1}{\sqrt{2\pi(w^2 + \sigma_y^2)}} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \\
&\times \frac{1}{\sqrt{2\pi\sigma_y^2 w^2/(w^2 + \sigma_y^2)}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\times (a_f n + cy) dy \\
&= \frac{1}{\sqrt{2\pi(w^2 + \sigma_y^2)}} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \left(a_f n + \frac{cw^2\mu}{w^2 + \sigma_y^2}\right)
\end{aligned}$$

If we substitute $\gamma = 1/(w^2 + \sigma_y^2)$ and $\mu = \bar{y}$, then,

$$\bar{W}(n, \bar{y}) = \sqrt{\frac{\gamma}{2\pi}} \exp\left(-\frac{\gamma\bar{y}^2}{2}\right) (a_f n + \gamma cw^2 \bar{y})$$

If we write $\bar{y} = \bar{a} + \bar{b}n$, then

$$\begin{aligned}
\beta &= \left(\frac{\partial/\partial\bar{a}}{\partial/\partial\bar{b}}\right) \ln \bar{W}(n, \bar{y}) \\
&= \left(\frac{\partial/\partial\bar{a}}{\partial/\partial\bar{b}}\right) \left(\ln \frac{\gamma}{2\pi} - \frac{\gamma(\bar{a} + \bar{b}n)}{2} + \ln(a_f n + \gamma cw^2(\bar{a} + \bar{b}n))\right) \\
&= \left(\frac{-\gamma}{2} + \frac{\gamma cw^2}{a_f n + \gamma cw^2(\bar{a} + \bar{b}n)}\right) \\
&= \left(\frac{-\gamma n}{2} + \frac{\gamma cw^2 n}{a_f n + \gamma cw^2(\bar{a} + \bar{b}n)}\right)
\end{aligned}$$

- ⁵ Now, we want to calculate $E(\beta)$ but this is difficult... Instead, we can still solve for
⁶ $\beta = 0$ for a given value of $n > 0$.